

COMPARATIVE SUSTAINABILITY BENCHMARKING OF PASSENGER VEHICLE MANUFACTURERS IN INDIA

A BRSR-Based ESG Analysis of
Tata Motors, Maruti Suzuki, and
Mahindra & Mahindra



Chetan Raj
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EXECUTIVE SUMMARY

India's automobile manufacturing sector is a key driver of economic growth, employment, and industrial innovation, but it is also among the most resource-intensive industries, with high energy consumption, greenhouse gas emissions, water use, and complex social and governance challenges. Passenger vehicle manufacturing, in particular, involves energy-intensive processes such as metal forming, painting, and assembly, while also facing workforce, safety, and supply-chain risks. As regulatory scrutiny and investor expectations around ESG performance intensify, sustainability has evolved from a compliance requirement to a strategic determinant of competitiveness and risk management.

This report benchmarks the sustainability performance of **Tata Motors, Maruti Suzuki, and Mahindra & Mahindra** within the Indian passenger vehicle segment using the **Business Responsibility and Sustainability Report (BRSR)** framework mandated by SEBI. The analysis converts BRSR disclosures into **intensity-based and percentage indicators** to enable fair comparison across firms of different scale. Performance is assessed across environmental, social, and governance dimensions aligned with **BRSR Core Attributes**, using clearly justified benchmarks classified as good, acceptable, or poor. Where direct quantitative data are unavailable, transparent and consistently applied

proxy indicators are used. The scope is limited to manufacturing operations, excluding downstream and Scope-3 impacts due to control and reporting inconsistencies.

The comparative analysis indicates clear differentiation in sustainability performance. **Tata Motors emerges as the overall sustainability leader**, consistently outperforming peers on energy intensity, greenhouse gas emissions intensity, water efficiency, renewable energy adoption, and board-level governance. This performance reflects effective decoupling of environmental impact from economic output and stronger integration of sustainability into operational and strategic decision-making. **Mahindra & Mahindra demonstrates credible transition momentum**, particularly in areas of transparency, water recycling, renewable energy adoption, and governance, although performance remains uneven across environmental efficiency indicators. **Maruti Suzuki, despite its market leadership, underperforms on several intensity-based environmental and governance metrics**, indicating higher transition risk and slower integration of sustainability into its scale-driven operating model.

Across all three companies, **workforce gender diversity remains a structural weakness**, with female participation well below acceptable benchmarks, highlighting a sector-wide inclusion

challenge rather than firm-specific underperformance. In contrast, workplace safety and training systems are relatively strong across the sector, suggesting maturity in core operational social practices. Governance performance varies significantly, with Tata Motors and Mahindra & Mahindra exceeding best-practice thresholds for board diversity, while Maruti Suzuki remains at minimum compliance levels.

Overall, the findings demonstrate that **sustainability leadership in India's passenger vehicle manufacturing sector is driven by efficiency, governance quality, and strategic intent rather than production scale.**

Intensity-based benchmarking provides a more decision-useful assessment of sustainability performance than absolute footprint measures and highlights the importance of integrating sustainability into core operations and governance structures. The analysis underscores the potential of BRSR, when applied through a structured benchmarking framework, to support informed corporate strategy, investor assessment, and policy design in India's evolving ESG landscape.

INTRODUCTION

India's automobile manufacturing sector fuels economic growth via industrial output, employment, and innovation, yet ranks among the most resource-intensive industries, with high energy use, GHG emissions, water consumption, and complex supply chains. Passenger vehicle production—encompassing energy-heavy processes like metal forming, painting, welding, and assembly—drives environmental depletion and social challenges, including workforce safety, skill gaps, gender diversity, and supplier oversight. Sustainability has evolved from peripheral duty to strategic necessity for competitiveness and risk management.

Globally, leaders adopt intensity metrics, renewables, water circularity, and robust governance to decouple growth from impacts. In India, SEBI's mandatory Business Responsibility and Sustainability Report (BRSR) for top listed firms—anchored in NGRBC—standardizes ESG

disclosures on energy, emissions, water, waste, diversity, safety, and governance. BRSR Core's quantitative focus aligns with global standards, amid rising investor demands for ESG-driven capital allocation.

This report benchmarks sustainability in the passenger vehicle segment using BRSR data from Tata Motors, Maruti Suzuki, and Mahindra & Mahindra. Benchmarking transforms raw disclosures into intensity-based metrics for fair comparisons, identifying leaders/ laggards, gaps in energy efficiency, emissions, water stewardship, and inclusion, and guiding strategy/policy. BRSR ensures credible, comparable, holistic analysis grounded in public data.

Scope limits to manufacturing operations (energy, emissions, water, waste, workforce, governance), excluding downstream impacts due to control/reporting inconsistencies.



Car manufacturing process (Image source: pexels.com)

INDUSTRY OVERVIEW

2.1 Environmental & Social Risk Profile

Passenger-vehicle manufacturing is energy-intensive (stamping, painting, heat treatment). Investor implications: Energy price volatility erodes margins; fossil-heavy grid exposure amplifies carbon pricing and divestment risks. Material metric(s): Energy (GJ) and Energy Intensity (GJ / ₹cr). Alpha signal: Renewable sourcing, energy efficiency programs, and process electrification drive cost leadership.

Manufacturing sites emit Scope 1 (fuel combustion) and Scope 2 (purchased electricity) GHGs; Scope-3 (vehicle use, supply chain) dominates lifecycle but is often inconsistently disclosed. Investor implications: Climate targets trigger regulatory fines and EPS pressure from activist investors. Operational levers: onsite renewables, grid-contracted RE, energy efficiency, and process electrification. Material metric(s): Scope 1+2 (tCO₂e), GHG Intensity (tCO₂e / ₹cr).

Many OEM plants are located in water-stressed regions (northern & western India). Investor implications:

Production halts from water shortages inflate capex; community/regulatory pushback delays projects. Material metric(s): Water withdrawal (kL), Water Intensity (kL / ₹cr), % recycled/reused.

The auto manufacturing workforce is large, unionised in places, and male-dominated. Investor implications: EV/software skills gaps drive attrition costs; gender imbalance hurts talent pipelines and institutional ESG screens. Material metric(s): % women in workforce, LTIFR, training coverage (% employees trained), wage equity proxies.

NIC-29 OEMs face material downside risks in energy/emissions (margin erosion), water (site continuity), supply-chain/materials (rare metals/semiconductors), and workforce (skills/safety). These lenses frame our benchmarking for risk-adjusted returns.

2.2 ESG Expectations for Auto OEMs

National & Global Benchmarks

India: SEBI's BRSR mandates quantitative disclosure for listed firms. NDC targets + BEE PAT/state water rules set compliance floors. Global: SBTi/Net-Zero alignment, RE100, credible Scope-3 now table stakes for

institutional capital. Key for investors: Intensity metrics (per ₹cr or per vehicle) reveal relative performance vs. absolute greenwashing.

Electric Vehicle (EV) Transition Pressures

Strategic pivot: Product portfolios, supply chains, factory footprints must shift to BEV/battery production. Capex implication: Battery assembly skills, power electronics, battery EOL waste,

surging electricity demand. Investment thesis: Early movers de-risk transition, capture market share premium, and generate superior ROIC.

Supply-Chain Risks

Geopolitical exposure : Semiconductors, rare earths, battery raw materials face chronic shortages. ESG contagion risk: Supplier violations trigger OEM fines, recalls, share price

volatility. Due diligence expectation: Robust supplier audits, corrective actions, capacity building now mandatory for fund approvals.



Supply Chain process (Image source: pexels.com)

COMPANY PROFILES

3.1 Tata Motors Limited

Tata Motors Limited is one of India's largest automobile manufacturers, operating across passenger vehicles, commercial vehicles, and electric mobility segments. In the passenger vehicle space, the company has a strong presence across internal combustion engine (ICE) vehicles as well as electric vehicles (EVs), with an expanding domestic market share. Tata Motors operates as part of the Tata Group and is listed on Indian stock exchanges, making it subject to SEBI's mandatory BRSR disclosure requirements.

The company's business model integrates vehicle design, manufacturing, and domestic sales, with increasing strategic emphasis on electrification and platform consolidation. Passenger vehicles form a significant component of Tata Motors' sustainability footprint due to the scale and energy intensity of manufacturing operations.



3.2 Maruti Suzuki India Limited

Maruti Suzuki India Limited is India's largest passenger vehicle manufacturer by production and sales volume. The company primarily focuses on mass-market passenger vehicles and operates as a subsidiary of Suzuki Motor Corporation. Maruti Suzuki's business model emphasizes high-volume manufacturing, cost efficiency, and extensive domestic market penetration.

Given its scale of operations, Maruti Suzuki's manufacturing activities represent a substantial share of the passenger vehicle sector's aggregate environmental footprint in India.



3.3 Mahindra & Mahindra Limited

Mahindra & Mahindra Limited (M&M) is a diversified automotive manufacturer with a strong presence in utility vehicles, SUVs, and electric mobility segments. In the passenger vehicle category, the company focuses on higher-value vehicles with growing emphasis on electric and alternative mobility solutions.

As part of the Mahindra Group, the company operates within a broader sustainability-oriented corporate framework and is subject to mandatory ESG disclosures under SEBI regulations.



Mahindra

METHODOLOGY & ANALYTICAL FRAMEWORK

This section outlines the methodological approach adopted to benchmark the sustainability performance of selected passenger vehicle manufacturers. The framework is designed to convert regulatory sustainability disclosures into **comparable, decision-useful indicators**, consistent with professional ESG benchmarking and sustainability consulting practices. Emphasis is placed on transparency, comparability, and policy alignment, in line with the intent of the **BRSR Core framework**.

4.1 Data Sources

BRSR Disclosures as the Primary Data Source

The analysis relies exclusively on **Business Responsibility and Sustainability Report (BRSR)** disclosures published by Tata Motors, Maruti Suzuki, and Mahindra & Mahindra. BRSR is mandated by the Securities and Exchange Board of India (SEBI) for listed entities and represents the most standardized and regulator-endorsed ESG disclosure framework in the Indian corporate context.

Using BRSR as the sole data source ensures:

- **Consistency** across companies operating under the same reporting framework
- **Credibility**, as disclosures are subject to regulatory scrutiny
- **Comparability**, particularly through the BRSR Core quantitative attributes

No third-party ESG ratings or proprietary databases are used, in order to avoid methodological opacity and rating-agency bias.

Financial Year Reference

All indicators are calculated using data from the **most recent financial year** reported in the respective BRSR documents. A single-year cross-sectional approach is adopted to ensure methodological consistency and to avoid distortions arising from changes in reporting boundaries or indicator definitions across years.

Publicly Available Reports

Only **publicly available corporate disclosures** are used, including:

- BRSR sections of annual or integrated reports
- Accompanying sustainability disclosures referenced within BRSR

No internal, unpublished, or estimated data are included. This approach aligns the analysis with investor-style ESG assessments that rely strictly on public information.

4.2 Indicator Selection Logic

Environmental, Social, and Governance Dimensions

Indicators are selected across **Environmental (E), Social (S), and Governance (G)** dimensions to reflect the multidimensional nature of sustainability risks in automobile manufacturing. Selection is guided by two principles:

1. **Sector materiality** for passenger vehicle manufacturing
2. **Data availability and consistency** within BRSR disclosures

Environmental indicators focus on resource efficiency and emissions intensity, social indicators address workforce resilience and inclusion, and governance indicators capture leadership accountability and value-chain oversight.

Alignment with BRSR Core Attributes

Indicator selection is explicitly aligned with **BRSR Core Attributes**, which emphasize:

- Quantitative measurement
- Outcome-oriented disclosures
- Comparability across firms

Priority is given to indicators that directly correspond to BRSR Core requirements (e.g., energy consumption, emissions, water use, workforce composition, board diversity). Where Core indicators are supplemented by additional disclosures, alignment with the **National Guidelines on Responsible Business Conduct (NGRBC)** is maintained.

This ensures that the analytical framework does not extend beyond the regulatory intent of BRSR, while still enabling meaningful benchmarking.

4.3 Indicator Construction

Intensity Metrics

To account for differences in firm size, production scale, and revenue, **intensity-based indicators** are used wherever feasible. Intensity metrics express environmental impact relative to economic output and are calculated as:

- Energy Intensity = Total Energy Consumption / Revenue

- GHG Emissions Intensity = Scope 1 + 2 Emissions / Revenue
- Water Intensity = Total Water Withdrawal / Revenue

These metrics enable **scale-neutral comparison** and reflect the efficiency with which firms convert resources into economic value.

Percentage-Based Metrics

For indicators related to composition or coverage, **percentage metrics** are used. These include:

- Renewable energy as a share of total energy
- Women employees as a percentage of total workforce
- Women directors as a percentage of board size

Percentage metrics are particularly suitable for social and governance indicators, where proportional representation is more informative than absolute numbers.

Proxy Indicators

In several areas, BRSR disclosures are qualitative or lack standardized quantitative measures. To address this limitation without fabricating data, **proxy indicators** are introduced. Proxies are used only when:

- The underlying sustainability issue is material
- Comparable quantitative data are unavailable across all firms

Examples include:

- **Training coverage** as a proxy for training intensity
- **Hazardous waste share** as a proxy for waste-related environmental risk
- **Supplier ESG oversight** as a proxy for value-chain sustainability governance

All proxy indicators are clearly identified as such and interpreted cautiously, focusing on **relative performance rather than precise measurement**.

4.4 Benchmark Development

Good / Acceptable / Poor Thresholds

Each indicator is evaluated against **three performance categories**:

- **Good** – Reflects leadership or best-practice performance
- **Acceptable** – Indicates compliance or moderate performance
- **Poor** – Signals underperformance or elevated sustainability risk

This categorical approach enables intuitive comparison and avoids false

precision in ESG assessment.

Policy and Literature Justification

Benchmark thresholds are not arbitrarily defined. They are justified using:

- Indian policy targets (e.g., BEE energy efficiency norms, national climate commitments)
- Labour participation and safety statistics
- Global ESG and sustainability literature (e.g., SBTi, ILO, OECD guidance)

Where sector-specific thresholds are unavailable, benchmarks are derived from **widely accepted best-practice ranges** used in sustainability consulting and ESG analysis.

Benchmark Table

Indicator	Good	Acceptable	Poor
Energy Intensity (GJ/₹cr)	< 10	10–25	> 25
Renewable Energy Share (%)	≥ 30%	10–30%	< 10%
GHG Intensity (tCO ₂ e/₹cr)	< 1.0	1.0–2.5	> 2.5
Water Intensity (kL/₹cr)	< 12	12–20	> 20
Women in Workforce (%)	≥ 30%	15–30%	< 15%

4.5 Scoring and Normalisation

Scoring Scale

Each indicator is assigned a numerical score based on benchmark category:

- **Good = 3**
- **Acceptable = 2**
- **Poor = 1**

This scoring system balances simplicity with analytical clarity and avoids over-weighting any single indicator.

Composite Indices

Scores are aggregated to create:

- **Environmental Score**
- **Social Score**
- **Governance Score**
- **Overall ESG Score**

Equal weighting is applied within each dimension to avoid subjective prioritization, unless otherwise justified. Composite indices are used solely for **relative comparison**, not as absolute measures of sustainability performance.

SUSTAINABILITY INDICATORS & BENCHMARKS

This section defines the sustainability indicators used for benchmarking passenger vehicle manufacturers and the associated performance benchmarks. Indicators are aligned with BRSR Core attributes, sectoral risk exposure, and established ESG practices. To ensure comparability across firms of different scale, intensity-based and percentage indicators are preferred over absolute values.

5.1 Environmental Indicators

Energy Intensity (GJ / ₹ crore)

Definition: Total energy consumed divided by revenue.

Benchmark: Good <10 | Acceptable 10–25 | Poor >25

Rationale: Energy efficiency is a core expectation under BEE and BRSR Core.

Intensity metrics enable scale-neutral comparison.

Renewable Energy Share (%)

Definition: Renewable energy as a percentage of total energy.

Benchmark: Good ≥30% | Acceptable 10–30% | Poor <10%

Rationale: Aligned with India's NDC targets and global clean-energy transition expectations.

GHG Emissions Intensity (tCO₂e / ₹ crore)

Definition: Scope 1 + 2 emissions divided by revenue.

Benchmark: Good <1.0 | Acceptable 1.0–2.5 | Poor >2.5

Rationale: Reflects climate impact relative to economic output and aligns with SBTi and TCFD guidance.

Water Intensity (kL / ₹ crore)

Definition: Total water withdrawal divided by revenue.

Benchmark: Good <12 | Acceptable 12–20 | Poor >20

Rationale: Manufacturing plants operate in water-stressed regions; efficiency is critical for operational continuity.

Water Recycling / Reuse (% – Proxy)

Definition: Recycled or reused water as a percentage of total withdrawal.

Benchmark: Good ≥40% | Acceptable 20–40% | Poor <20%

Rationale: CPCB guidelines encourage high levels of industrial water reuse.

Hazardous Waste Share (% – Proxy)

Definition: Hazardous waste as a percentage of total waste generated.

Benchmark: Good <15% | Acceptable 15–30% | Poor >30%

Rationale: Lower hazardous share indicates cleaner production processes.

5.2 Social Indicators

Gender Diversity (% Women Employees)

Definition: Women employees as a percentage of total workforce.

Benchmark: Good $\geq 30\%$ | Acceptable 15–30% | Poor $< 15\%$

Rationale: Benchmarked against national female labour participation and SDG 5.

Training Coverage (% – Proxy)

Definition: Employees trained as a percentage of total workforce.

Benchmark: Good $\geq 90\%$ | Acceptable 60–90% | Poor $< 60\%$

Rationale: Continuous upskilling is critical due to EV and automation transition.

Workplace Safety (LTIFR)

Definition: Lost Time Injury Frequency Rate as reported.

Benchmark: Good $= 0.00$ | Acceptable 0.01–0.10 | Poor > 0.10

Rationale: Manufacturing best practice targets zero harm.

5.3 Governance Indicators

Board Gender Diversity (%)

Definition: Women directors as a percentage of board size.

Benchmark: Good $\geq 30\%$ | Acceptable 10–30% | Poor $< 10\%$

Rationale: SEBI LODR mandates representation; governance literature identifies 30% as critical mass.

Supplier ESG Oversight (Proxy)

Definition: Qualitative assessment of supplier ESG policies and audits.

Benchmark: Good = policy + audits | Acceptable = policy only | Poor = no disclosure

Rationale: Automotive supply chains carry high ESG risk, making oversight material.

The benchmark framework converts BRSR disclosures into decision-useful performance categories. Environmental indicators capture transition readiness, social indicators reflect workforce resilience, and governance indicators assess accountability, forming the analytical base for comparative results.

COMPARATIVE RESULTS & ANALYSIS

This section presents a comparative analysis of Tata Motors, Maruti Suzuki, and Mahindra & Mahindra based strictly on Business Responsibility and Sustainability Report (BRSR) disclosures. The analysis aligns with BRSR Core Attributes and uses intensity-based indicators and justified proxies to ensure comparability across firms of different scales.

6.1 Environmental Performance

Energy Consumption and Intensity

Tata Motors reports total energy consumption of 2,940,132 GJ with an energy intensity of 6.71 GJ per ₹ crore of revenue. Maruti Suzuki reports the highest absolute energy consumption at 5,086,820 GJ and the highest energy intensity at 37.69 GJ per ₹ crore. Mahindra & Mahindra reports 2,878,986 GJ with an energy intensity of 20.36 GJ per ₹ crore. This indicates Tata Motors' superior energy efficiency relative to revenue.

Company	Total Energy Consumption (GJ)	Revenue (₹ crore)	Energy Intensity (GJ / ₹ crore)
Tata Motors	2,940,132	437,928	6.71
Maruti Suzuki	5,086,820	134,937	37.69
Mahindra & Mahindra	2,878,986	141,373	20.36

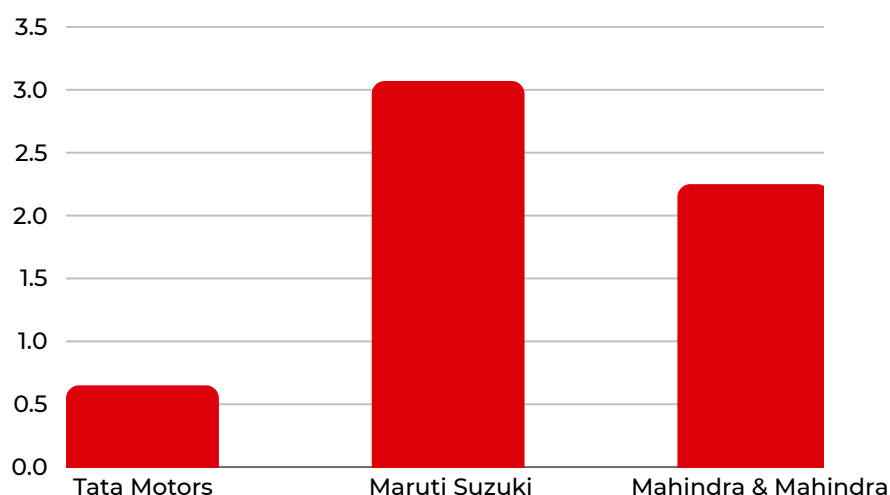
Renewable Energy Adoption

Tata Motors derives approximately 30.48% of its total energy from renewable sources, compared to 18.47% for Mahindra & Mahindra and 6.07% for Maruti Suzuki. This reflects stronger alignment of Tata Motors with clean energy transition objectives.

Company	Renewable Energy (GJ)	Renewable Share (%)
Tata Motors	896,271	30.48%
Maruti Suzuki	~308,000	6.07%
Mahindra & Mahindra	~531,000	18.47%

GHG Emissions (Scope 1 and 2)

Tata Motors reports combined Scope 1 and 2 emissions of 285,035 tCO₂e, translating to an emissions intensity of 0.65 tCO₂e per ₹ crore. Mahindra & Mahindra reports 317,659 tCO₂e (2.25 intensity), while Maruti Suzuki reports 414,612 tCO₂e (3.07 intensity). Tata Motors therefore demonstrates the strongest decoupling of emissions from economic output.



Water Use and Recycling

Tata Motors reports total water withdrawal of 4,492,008 kL and the lowest water intensity (10.26 kL per ₹ crore). Maruti Suzuki reports 1,946,609 kL (14.42 intensity), while Mahindra & Mahindra reports 2,880,263 kL with the highest water intensity (20.38). Mahindra & Mahindra is the only firm to explicitly report water recycling at 39%, indicating higher transparency.

Company	Total Water Withdrawal (kL)	Water Intensity (kL / ₹ crore)	Water Recycled (%)
Tata Motors	4,492,008	10.26	Not explicitly reported
Maruti Suzuki	1,946,609	14.42	Not explicitly reported
Mahindra & Mahindra	2,880,263	20.38	39%

Hazardous Waste Management

Mahindra & Mahindra provides consolidated hazardous waste figures, whereas Tata Motors and Maruti Suzuki disclose waste by category without consolidated totals. This limits comparability but indicates stronger reporting clarity by Mahindra & Mahindra.

6.2 Social Performance

Gender Diversity

Female workforce participation remains low across all firms. Tata Motors reports approximately 9.9% women employees, Mahindra & Mahindra about 11.5%, and Maruti Suzuki 5.85%. This highlights a sector-wide structural inclusion gap.

Company	Total Employees	Women Employees	Women (%)
Tata Motors	12,956	1,280	~9.9%
Maruti Suzuki	14,040	822	5.85%
Mahindra & Mahindra	23,197	2,665	~11.5%

Training and Human Capital Development (Proxy)

All three companies disclose extensive training programs but do not report average training hours per employee. Using training coverage and program disclosures as proxies, Tata Motors and Mahindra & Mahindra demonstrate stronger human capital development systems than Maruti Suzuki.

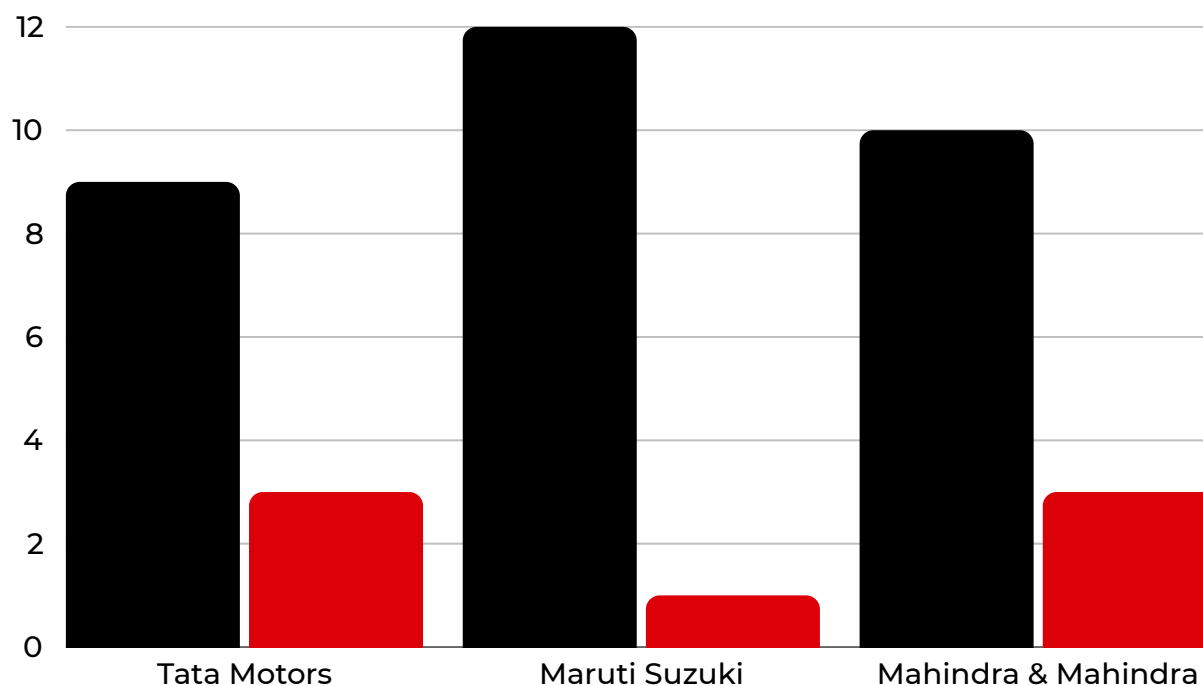
Occupational Health and Safety

Employee LTIFR is reported at approximately 0.09 for Tata Motors, while both Maruti Suzuki and Mahindra & Mahindra report zero LTIFR. Overall, safety performance is strong across all firms.

6.3 Governance Performance

Board Gender Diversity

Tata Motors reports 33.33% women directors, Mahindra & Mahindra 30%, and Maruti Suzuki 8.33%. Tata Motors and Mahindra & Mahindra exceed global best-practice thresholds, while Maruti Suzuki remains at minimum compliance levels.



Comparative Summary

Dimension	Leader	Middle	Laggard
Energy Efficiency	Tata Motors	Mahindra & Mahindra	Maruti Suzuki
Renewable Energy	Tata Motors	Mahindra & Mahindra	Maruti Suzuki
Carbon Intensity	Tata Motors	Mahindra & Mahindra	Maruti Suzuki
Water Efficiency	Tata Motors	Maruti Suzuki	Mahindra & Mahindra
Workforce Diversity	Mahindra & Mahindra	Tata Motors	Maruti Suzuki
Board Diversity	Tata Motors	Mahindra & Mahindra	Maruti Suzuki

BENCHMARK-BASED INSIGHTS

This section synthesises benchmarked sustainability results derived from BRSR disclosures to identify **patterns of leadership, structural gaps, and strategic trade-offs** within India's passenger vehicle manufacturing sector. The discussion focuses on **relative performance and underlying drivers**, rather than reiterating numerical results, in line with the intent of the BRSR Core framework.

7.1 Leaders vs Laggards: Who Performs Best and Why

Benchmark-based analysis reveals that **sustainability leadership is not correlated with market dominance**, but with the depth of strategic integration and governance quality.

Tata Motors emerges as the strongest overall sustainability performer. Its consistent outperformance on intensity-based environmental indicators reflects a deliberate effort to decouple environmental impact from economic activity. This suggests that sustainability considerations are embedded within operational planning and capital allocation, rather than treated as post-hoc compliance. Such decoupling aligns directly with the BRSR Core emphasis on efficiency, resilience, and transition readiness. Strong board-level gender diversity further reinforces Tata Motors' governance maturity, indicating that sustainability oversight is present at the highest decision-making levels.

Mahindra & Mahindra demonstrates credible transition momentum, particularly on indicators related to transparency and resource circularity. Explicit disclosure of water recycling performance and comparatively stronger renewable energy adoption point to a pragmatic sustainability strategy focused on risk mitigation and operational resilience. Mahindra also performs relatively better on social indicators, suggesting alignment with inclusive growth objectives, although absolute performance levels remain moderate. Overall, Mahindra represents a firm in transition—strategically moving toward sustainability leadership, but not yet consistently outperforming across all dimensions.

In contrast, **Maruti Suzuki consistently appears as the relative laggard** in benchmarked sustainability performance. Despite its dominant position in passenger vehicle production, Maruti underperforms on several efficiency-based environmental indicators and governance diversity benchmarks. This suggests that its historically successful cost- and volume-driven business model has not been accompanied by equivalent sustainability integration. Legacy infrastructure, energy-intensive processes, and slower organisational adaptation appear to constrain performance when evaluated through intensity-based metrics.

The leaders-versus-laggards comparison reinforces a central insight: **sustainability leadership is driven by strategic intent and governance capability, not by scale alone.**

7.2 Structural Gaps: Sector-Wide Sustainability Challenges

Beyond firm-level differences, the analysis highlights **structural sustainability gaps that persist across the passenger vehicle manufacturing sector.**

The most significant gap is **gender inclusion in the workforce.** All three companies report low female participation, indicating a systemic issue rather than isolated underperformance. This reflects deeper structural constraints, including shop-floor job design, limited female participation in technical talent pipelines, and entrenched cultural norms within manufacturing environments. The persistence of this gap—even among relatively strong ESG performers—suggests that firm-level initiatives alone may be insufficient without coordinated sectoral and policy support.

A second structural challenge is **uneven depth of sustainability disclosure.** While BRSR has improved transparency, companies differ in how comprehensively they quantify outcomes, particularly for social and waste-related indicators. Several metrics are disclosed narratively rather than numerically, limiting analytical comparability. This highlights a broader ESG reporting limitation: compliance with disclosure requirements does not automatically ensure data usefulness for benchmarking and decision-making.

A third gap relates to **environmental transition depth.** Efficiency improvements are more visible than progress in renewable energy integration or circular resource use. This suggests that firms prioritise sustainability actions with clearer short-term efficiency benefits, while more capital-intensive transitions progress unevenly. As a result, sector-wide sustainability performance remains constrained by both economic and technological factors.

Collectively, these gaps indicate that **sector-level sustainability outcomes are shaped as much by industry structure and labour-market realities as by individual corporate strategies.**

7.3 Trade-offs in Sustainability Performance

Scale vs Efficiency

A key insight from the benchmark-based assessment is the **trade-off between operational scale and sustainability efficiency.** Large manufacturers benefit from economies of scale and cost advantages, but these same characteristics

can create inertia, making it harder to rapidly redesign production systems or transition to cleaner energy sources.

Intensity-based metrics reveal that firms with stronger sustainability performance are not necessarily the largest producers, but those that prioritise **process optimisation and governance-led decision-making**. As regulatory and investor focus increasingly shifts toward efficiency and transition-aligned metrics, reliance on scale alone may expose firms to rising sustainability risks.

Cost vs Sustainability

The analysis also highlights a trade-off between **short-term cost optimisation and long-term sustainability investment**. Investments in renewable energy, water recycling, workforce diversification, and governance systems often involve upfront costs without immediate financial returns. Firms focused on cost leadership may therefore delay such investments.

However, benchmarked outcomes suggest that early sustainability investments reduce long-term intensity risks, strengthen governance credibility, and improve resilience to regulatory tightening. In this sense, sustainability functions as a **risk-management and value-preservation strategy**, consistent with the BRSR Core philosophy.

Key Takeaway

Overall, the benchmark-based discussion demonstrates that **sustainability performance in India's passenger vehicle sector is driven more by strategic integration, governance maturity, and efficiency orientation than by production scale or market dominance**. Tata Motors currently leads on most benchmarked dimensions, Mahindra & Mahindra shows credible transition potential, and Maruti Suzuki faces clear sustainability catch-up challenges. Persistent structural gaps—particularly in gender inclusion and disclosure depth—highlight the limits of firm-level action and the need for broader sectoral and policy interventions.

STRATEGIC IMPLICATIONS

This section translates benchmark-based findings into **targeted strategic actions** for individual firms and the passenger vehicle sector. Recommendations focus on **material leverage points** where sustainability performance can be improved without undermining commercial competitiveness, in line with the BRSR Core framework.

8.1 Company-Specific Recommendations

Tata Motors: Consolidate and Signal Leadership

Strategic implication:

Tata Motors operates at a relatively advanced stage of sustainability maturity. The priority is to **consolidate leadership and deepen system-wide integration**.

Key recommendations:

- **Move from efficiency leadership to transition leadership** by linking energy and emissions efficiency gains to clear medium-term transition targets.
- **Extend sustainability integration into the value chain**, particularly through supplier capability-building on energy and emissions management.
- **Strengthen leadership pipeline diversity**, ensuring board-level inclusiveness translates into senior management and technical roles.
- **Leverage sustainability as a strategic differentiator** in investor and product positioning, especially in electric and low-emission segments.

Maruti Suzuki: Shift from Scale Efficiency to Sustainability Efficiency

Strategic implication:

Maruti Suzuki's sustainability challenge lies in aligning its scale-driven model with **efficiency-based sustainability outcomes**.

Key recommendations:

- **Prioritise intensity-based improvements** in energy, emissions, and water use rather than focusing on absolute footprint narratives.
- **Accelerate renewable energy adoption** through on-site generation and long-term green power procurement at manufacturing locations.
- **Enhance governance diversity beyond minimum compliance** to strengthen strategic oversight of sustainability risks.
- **Embed sustainability targets into operational KPIs**, ensuring accountability at plant and managerial levels.

Mahindra & Mahindra: Convert Transition Momentum into Consistency

Strategic implication:

Mahindra & Mahindra shows strong transition intent but uneven performance. The focus should be on **standardisation and scale-up**.

Key recommendations:

- **Standardise best practices across plants** using internal benchmarking on energy, water, and emissions performance.
- **Deepen circular economy initiatives**, extending beyond water reuse to materials and waste management.
- **Improve quantitative disclosure of social indicators** to strengthen benchmarking credibility.
- **Align inclusion and skilling initiatives with future workforce needs**, positioning social performance as a productivity enabler.

8.2 Sector-Level Recommendations

Policy Alignment

- **Link incentives to intensity-based improvements**, encouraging efficiency rather than scale expansion.
- **Address gender inclusion structurally** through targeted skilling, safety standards, and inclusive manufacturing design.
- **Progressively standardise BRSR metrics**, especially for social and waste indicators, to improve comparability.

Industry Best Practices

- **Promote collaborative platforms** for renewable energy sourcing, circular practices, and supplier ESG management.
- **Develop sector-specific sustainability benchmarks** to guide performance convergence.
- **Create shared skilling ecosystems** to support technological transition and inclusive workforce participation.

LIMITATIONS & USE OF PROXIES

This analysis is subject to **several data-related limitations** inherent in the current structure and implementation of Business Responsibility and Sustainability Reporting (BRSR) disclosures. While core environmental indicators such as energy consumption, greenhouse gas emissions, and water withdrawal are generally reported in a consistent and quantifiable manner, **several socially and operationally relevant variables are either partially disclosed or presented in non-standardised formats**. In particular, none of the analysed companies provide a consolidated numerical figure for average training hours per employee, despite extensive narrative descriptions of training programmes and learning initiatives. Similarly, wage-disaggregated indicators, such as the proportion of wages paid to women, are not uniformly disclosed, limiting the ability to assess depth of economic inclusion and human capital investment across firms.

Comparability is further **constrained by differences in reporting boundaries and disclosure practices**. Tata Motors reports several indicators on a consolidated group basis, while Maruti Suzuki largely follows a standalone reporting approach, with Mahindra & Mahindra adopting mixed boundaries depending on the indicator. These variations affect the comparability of absolute values and necessitate a greater analytical reliance on intensity-based metrics rather than total quantities. Additionally, the level of quantification versus narrative disclosure varies across firms, particularly for social, training, waste, and supply-chain-related indicators. Although all companies comply with BRSR reporting requirements, this heterogeneity reduces the precision of cross-company benchmarking and reflects broader challenges in the current phase of ESG reporting standardisation in India.

To address these limitations without extrapolating or fabricating data, the analysis employs carefully **defined proxy indicators** where direct quantitative measures are unavailable. Training coverage is used as a proxy for training intensity, capturing the breadth of employee participation in skill development initiatives even though it does not reflect training duration or depth. Hazardous waste share is used as a proxy for waste-related environmental risk, based on the clarity and structure of hazardous waste disclosures relative to total waste streams, rather than on absolute quantities. Similarly, supplier ESG oversight is used as a proxy for supply-chain sustainability, inferred from disclosures relating to supplier codes of conduct, ESG audits, monitoring mechanisms, and corrective action processes. All proxies are applied consistently across companies, explicitly acknowledged as proxies, and interpreted qualitatively rather than as precise measurements. This approach ensures methodological transparency, analytical integrity, and reproducibility while operating within the constraints of available BRSR data.

CONCLUSION

This report set out to critically examine and benchmark the sustainability performance of selected **passenger vehicle manufacturers in India** using **Business Responsibility and Sustainability Reporting (BRSR)** disclosures as the primary analytical foundation. Anchored in the BRSR Core Attributes, the analysis moved beyond descriptive reporting to develop intensity-based indicators, construct justified benchmarks, and enable meaningful cross-company comparison. By focusing on **environmental efficiency, social inclusion, and governance quality**, the report demonstrated how BRSR data can be systematically translated into an analytically robust sustainability assessment framework that resembles professional ESG consulting practice rather than compliance-driven reporting.

The comparative findings reveal clear differentiation in sustainability performance across firms. **Tata Motors consistently emerges as the strongest performer** on efficiency-based environmental indicators and governance-related dimensions, reflecting a higher degree of strategic integration of sustainability into operational and decision-making processes. **Mahindra & Mahindra demonstrates credible transition momentum**, particularly in areas related to transparency, resource circularity, and social indicators, though performance remains uneven across dimensions. In contrast, **Maruti Suzuki, despite its dominant market position, underperforms on several benchmarked** sustainability indicators, highlighting a misalignment between scale-driven production efficiency and sustainability efficiency. These differences underscore that sustainability leadership is shaped more by strategic intent, governance maturity, and process optimisation than by production scale alone.

At the same time, the analysis highlights several structural challenges that transcend firm-level performance. Persistently low female workforce participation across all companies points to deep-rooted sectoral barriers within automobile manufacturing that cannot be addressed solely through individual corporate initiatives. Inconsistencies in disclosure depth and reporting boundaries further limit analytical comparability, particularly for social and supply-chain-related indicators. These constraints reflect the transitional nature of ESG reporting in India, where regulatory frameworks have significantly improved transparency but have not yet achieved full standardisation or analytical completeness.

A central takeaway from this study is the importance of intensity-based and benchmark-driven assessment in evaluating corporate sustainability performance. Absolute environmental footprints, while informative, provide

limited insight into efficiency, resilience, and transition readiness. In contrast, intensity metrics and carefully justified benchmarks allow for more meaningful comparison across firms of different sizes and business models. The use of proxy indicators, transparently acknowledged and consistently applied, further demonstrates how analytical rigour can be maintained even in the presence of incomplete data. This approach reinforces the value of methodological transparency in sustainability analysis and highlights the importance of explicitly communicating limitations rather than obscuring them.

The findings of this report have important implications for ESG reporting and decision-making. For corporate managers, the analysis illustrates how BRSR disclosures can be used not only to meet regulatory requirements but also to identify strategic weaknesses, prioritise sustainability investments, and track progress relative to peers. For investors and analysts, benchmark-based insights offer a more nuanced understanding of sustainability risk and opportunity than headline ESG scores or standalone indicators. For policymakers and regulators, the study highlights areas where further standardisation and guidance—particularly in social and value-chain disclosures—could enhance the effectiveness of the BRSR framework as a decision-useful reporting tool.

Overall, this report demonstrates that BRSR, when applied through a structured analytical framework, has the potential to function as a powerful instrument for sustainability benchmarking and strategic evaluation in the Indian corporate context. However, realising this potential requires a shift in both corporate practice and stakeholder expectations—from viewing sustainability reporting as an end in itself to treating it as a means of informed decision-making and long-term value creation. As ESG considerations become increasingly central to regulatory scrutiny, capital allocation, and corporate strategy, firms that integrate sustainability into core operations and governance structures will be better positioned to manage transition risks and sustain competitive advantage in a rapidly evolving business environment.

DECLARATION OF AI USE

This report was prepared with the limited use of artificial intelligence (AI)-based tools to support the research and writing process. AI tools were utilized primarily for **language refinement, structural organization, formatting assistance, and conceptual clarification** in order to enhance the clarity, coherence, and professional presentation of the report.

AI assistance was also used to support **analytical structuring**, including the development of comparative frameworks, benchmarking logic, and visualization concepts. However, all **data collection, selection of sustainability indicators, interpretation of Business Responsibility and Sustainability Report (BRSR) disclosures, benchmarking thresholds, comparative assessments, and final conclusions** were conducted by the author.

No AI tools were used to fabricate, manipulate, or alter primary data. All numerical values, classifications, and sustainability assessments are based on **publicly available BRSR disclosures and authoritative policy or ESG literature**, and responsibility for their accuracy rests solely with the author.

The use of AI was intended solely as a **supportive aid** and not as a substitute for independent academic judgment, critical analysis, or original work. The author confirms that this report complies with institutional guidelines on academic integrity and responsible use of AI tools.

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